LeoWay - System Specification

Version 0.4 · Draft for review

Team: LRD (Leo Research Department)

Authors: Laurenz (LPuehringerStudent), Ayan (Ayan2310), David (David-Fruehwirt), Behrang (BehrangEl11).

1. Initial situation and goal

1.1 Application domain

Most HTL Leonding students come by public transport: Linz AG bus 19 at Limesstraße, trams 3 and 4 and buses 11, 19 and 191 at Meixnerkreuzung, regional OÖVV buses such as 611 and 618, and ÖBB trains. Lessons start at 07:05 (period 0) or 08:00 (period 1).

The school plans lessons in WebUntis, including cancellations, substitutions and exams. Teachers record late arrivals there as a short absence; the form teacher decides whether it is excused. Linz AG publishes live departures; the other operators publish timetables. LeoWay connects the student's WebUntis data with recorded transport delays.

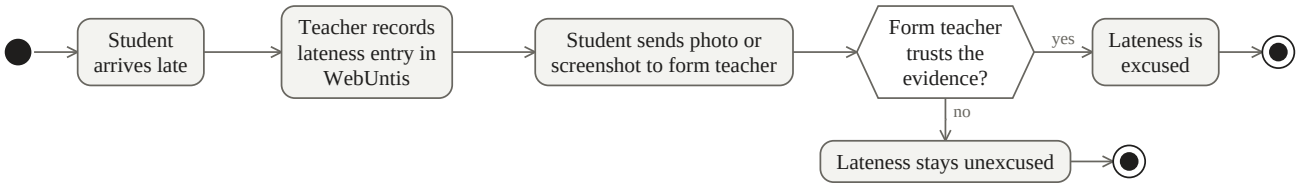
1.2 Terms

Term	Meaning
Access key	The student's WebUntis key (Profil → Freigaben), used instead of the password.
Form teacher	The class teacher (Klassenvorstand) who decides about excusing lateness.
First real lesson	The first lesson that still takes place after cancellations and changes.
Exam day	A day with an exam (e.g. a Schularbeit) from WebUntis. Manual marking is a fallback when exam data is missing.
Leave-by time	When to leave home to reach the first real lesson, including walking time and a buffer.
Lateness entry	A WebUntis absence starting at the first real lesson and ending within it; longer absences are not matched automatically.
Delay record	One reported state of a departure, with its line, trip, stop, bay, service date, planned time, reported delay, trip status, source and collection time. A departure gets a new record each time its delay or trip status changes. The latest record is the last prediction, not a measured departure.
Delay appeal	An email from the student to the form teacher that links a lateness entry to a delay record.
Verification link	A link in the appeal showing only the shared delay record.
Crowd report	A disruption reported by a student; several reports can trigger a warning.
auspex-lang	A small rule language for personal alerts, such as a delay over five minutes on an exam day.

1.3 Business processes today

BP1 · Getting to school. The student checks WebUntis for changes and, separately, Scotty or the Linz AG app for departures, and works out the leave-by time in their head. Delays are usually noticed at the stop, too late for an earlier connection.

BP2 · Excusing lateness. The teacher in the first lesson records a lateness entry in WebUntis. The student sends a photo of the vehicle display or a Scotty screenshot to the form teacher, who cannot check whether it is genuine or from that morning.



BP3 · Reporting problem connections. Nobody collects evidence about connections that are often late, so the student council has nothing to take to Linz AG or OÖVV.

1.4 Goal

Goal	Target
Timetable-aware alerts	Every alert uses the student's first real lesson, including cancellations, substitutions and exams (BP1).
Early delay warnings	A student whose saved journey will arrive late is warned and offered another connection where one exists (BP1).
Trustworthy appeals	A late student prepares an appeal in under one minute; the form teacher checks it through the verification link (BP2).
Data history	The delay recorder runs continuously from the first sprint, including holidays (BP3).
Crowd reports	A disruption confirmed by several students triggers a warning, often before official feeds show it.
Prediction	The model beats two baselines: "same delay as yesterday" and "average delay of this trip".
School statistics	One report on connections that regularly arrive after lessons start, handed to the student council (BP3).

Limits: LeoWay only reads WebUntis. Students send appeals themselves and teachers still decide; a delay record shows what a source reported, not that the student was on the vehicle. Not included: GPS, home addresses and a native iPhone app (iPhone users use the website). Other schools, indoor navigation and a public delay dataset are later extensions.

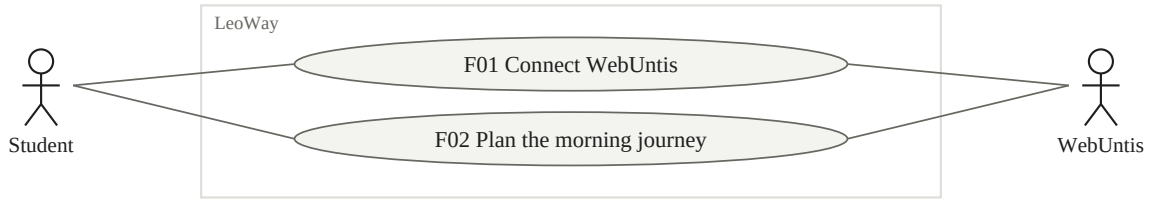
2. Functional requirements

Requirements are grouped into four blocks, each with a use case diagram. Use cases are numbered F01–F12 and each has its own acceptance test in section 6. Student use cases start from the home screen, called **Home** below:

Today	**Last sync 05:45**
First lesson	08:55
Leave by	07:42
Bus 19	+6 min
[View another connection]	[Report disruption]
[My lateness / draft appeal]	[Route and alerts]

2.1 Block A: Timetable and leave-by time

The student connects WebUntis once and saves a route; LeoWay then calculates the leave-by time every school day.



F01 · Connect WebUntis

Business process	BP1 Getting to school, BP2 Excusing lateness	**Goal**	LeoWay can read the student's lessons, exams and lateness entries.
Precondition	The student has a LeoWay account and a WebUntis access key.	**Postcondition**	The key is stored encrypted; the first sync has loaded the timetable. Absence reasons and notes are never stored.
Involved roles	Student; WebUntis (external system)	**Triggering event**	The student taps "Connect WebUntis" during setup or in Route and alerts.

Standard scenario. The student enters username and access key. LeoWay logs in, loads the timetable and shows the next first real lesson.

Non-standard scenario. The key is wrong or revoked: LeoWay stores nothing and shows "Access key not accepted" with help. If WebUntis is down, the student retries later.

Call: Route and alerts

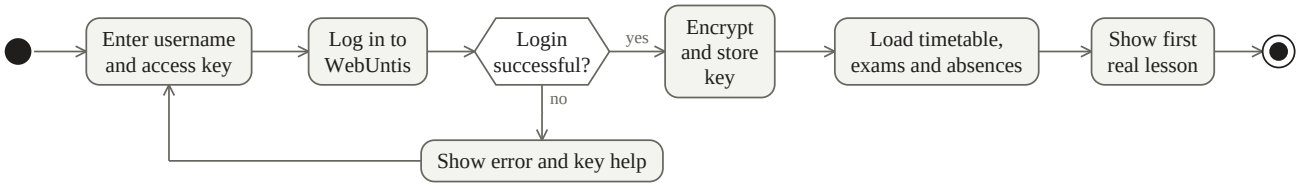
WebUntis not connected
[Connect WebUntis]

Standard: connected

Username [max.muster]
Access key [••••••••]
Connected · first lesson 08:00

Non-standard: key rejected

Access key [••••••••]
Error: access key not accepted
[How to create a key]



F02 · Plan the morning journey

Business process	BP1 Getting to school	**Goal**	The student knows when to leave home for the first real lesson.
Precondition	WebUntis is connected (F01).	**Postcondition**	A leave-by time exists for the next school day and is updated when lessons change.
Involved roles	Student; WebUntis	**Triggering event**	The student saves a route, or a timetable sync finds a changed first lesson.

Standard scenario. The student picks home and school stops and walking times; LeoWay suggests a buffer (proposed: 5 minutes) and shows the leave-by time for the latest fitting connection.

Non-standard scenario. The first lesson is cancelled overnight: the next sync moves the leave-by time and the phone alarm (F04). If no connection fits, LeoWay shows the earliest possible arrival.

Call: Route and alerts

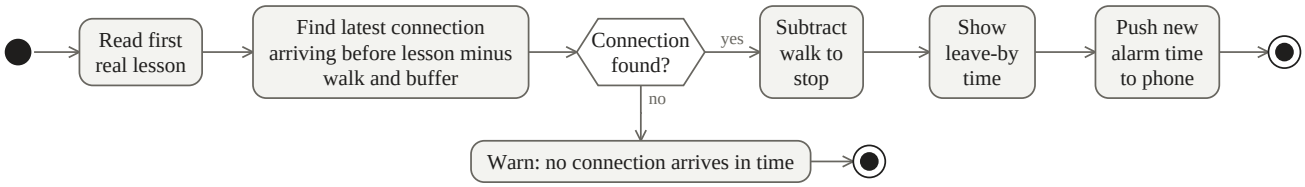
Home stop Leonding Zentrum
School stop Meixnerkreuzung
[Edit route]

Standard: leave-by time

First lesson 08:00
Bus 19 07:31 → 07:47
Leave by 07:24

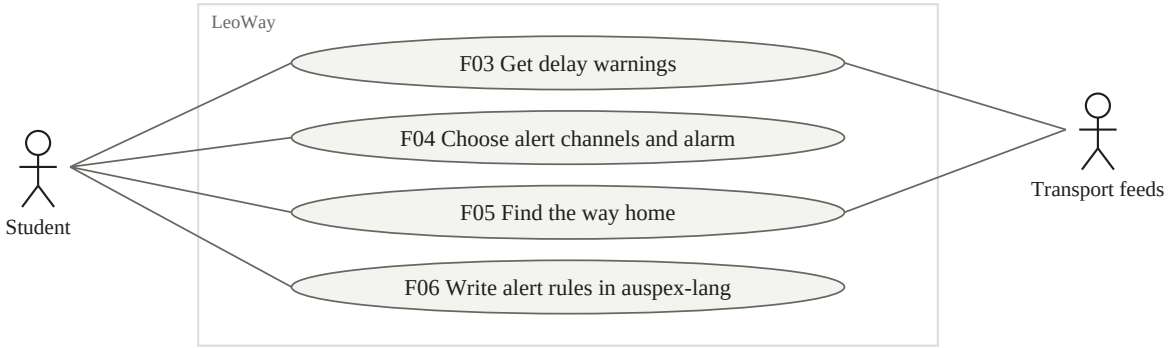
Non-standard: lesson cancelled

First lesson 08:55 (08:00 cancelled)
Leave by 08:16 (was 07:24)
Alarm moved



2.2 Block B: Alerts

LeoWay watches the student's connection and alerts them in the channels they chose.



F03 · Get delay warnings

Business process	BP1 Getting to school	**Goal**	The student learns early that the saved journey will arrive late and gets an alternative.
Precondition	A leave-by time exists (F02); at least one alert channel is set (F04).	**Postcondition**	The student has received one warning per channel for the change; the alternative is shown on Home.
Involved roles	Student; transport feeds	**Triggering event**	A feed reports a delay or cancellation on the student's connection.

Standard scenario. Bus 19 is +6 min, so the student would arrive 3 minutes late. LeoWay warns and suggests tram 3, adding an exam note if the first lesson is an exam.

Non-standard scenario. Without live data for the line, LeoWay shows "timetable only" and the last update time, and sends no delay warning.

Call: Home

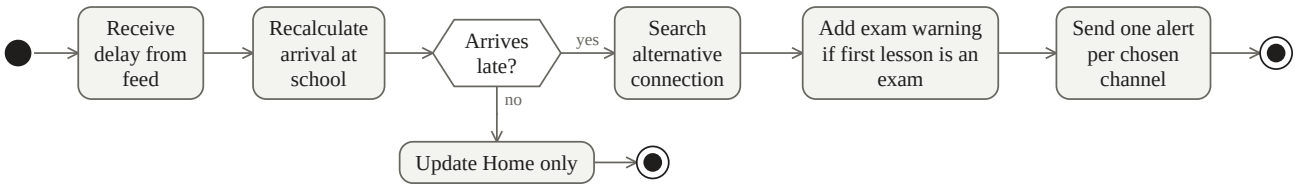
Bus 19 +6 min
[View another connection]

Standard: warning

Warning: Bus 19 +6 min, arrive 08:03
Exam in first lesson
Take tram 3 at 07:29 instead

Non-standard: no live data

Bus 611 07:10
Timetable only
Last update 05:41

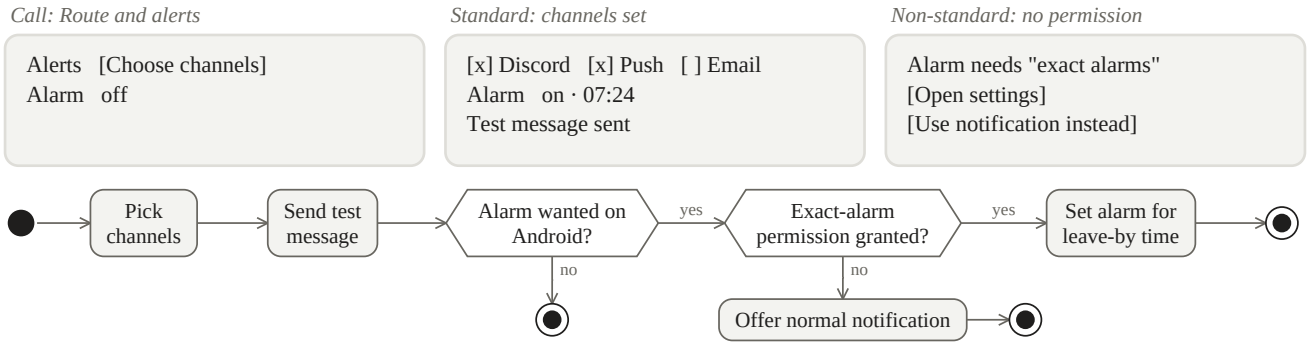


F04 · Choose alert channels and alarm

Business process	BP1 Getting to school	**Goal**	Alerts reach the student where they look; on Android, a leave-by alarm rings.
Precondition	The student is logged in. For the alarm: the Android app is installed.	**Postcondition**	Chosen channels are stored; the Android alarm is set for the current leave-by time.
Involved roles	Student; Discord, push service, email	**Triggering event**	The student opens "Route and alerts".

Standard scenario. The student picks Discord and push and gets a test message on each. On Android, the alarm is set for the leave-by time and follows its changes. Channels are built in the order Discord, push, email.

Non-standard scenario. Android denies the exact-alarm permission. The app explains why it is needed and offers a normal high-priority notification instead.

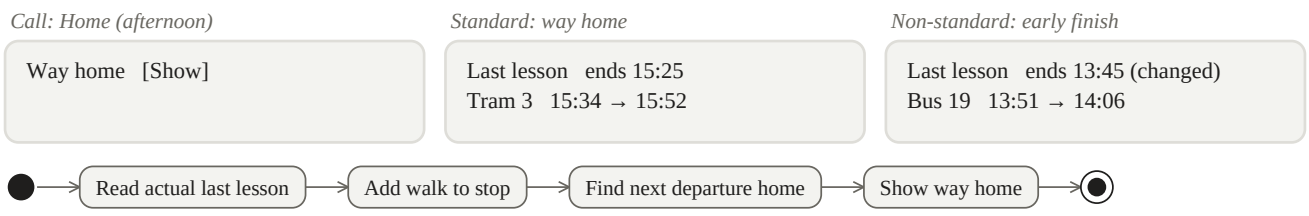


F05 · Find the way home

Business process	BP1 Getting to school (return trip)	**Goal**	The student sees the next connection home after the actual last lesson.
Precondition	WebUntis is connected and a route is saved.	**Postcondition**	Home shows the next journey home.
Involved roles	Student; WebUntis; transport feeds	**Triggering event**	The last lesson of the day ends, or a sync finds that lessons end early.

Standard scenario. The last lesson ends at 15:25. LeoWay shows tram 3 at 15:34, allowing the walk to the stop.

Non-standard scenario. The last two lessons are cancelled. The next sync moves the journey home to the new end time, 13:45, and shows the next connection after it.

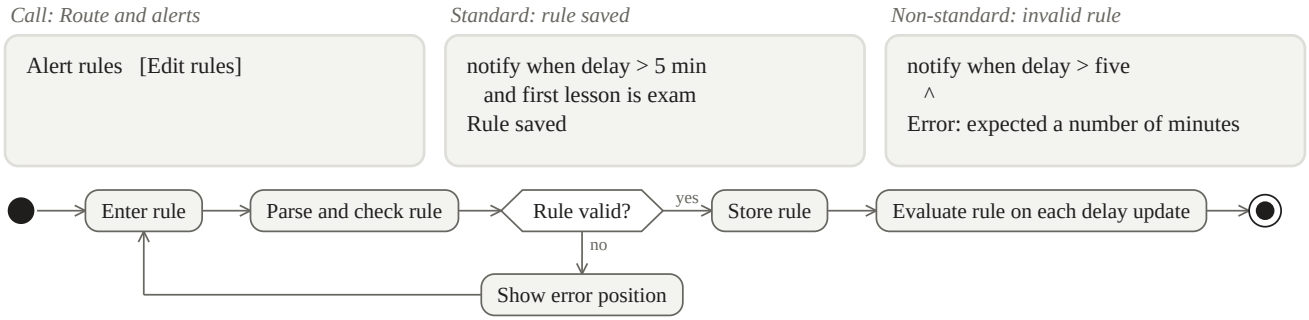


F06 · Write alert rules in auspex-lang

Business process	BP1 Getting to school	**Goal**	The student gets alerts only for situations that matter to them.
Precondition	WebUntis is connected; at least one channel is set (F04).	**Postcondition**	A valid rule is stored and checked with every delay update.
Involved roles	Student	**Triggering event**	The student opens "Alert rules".

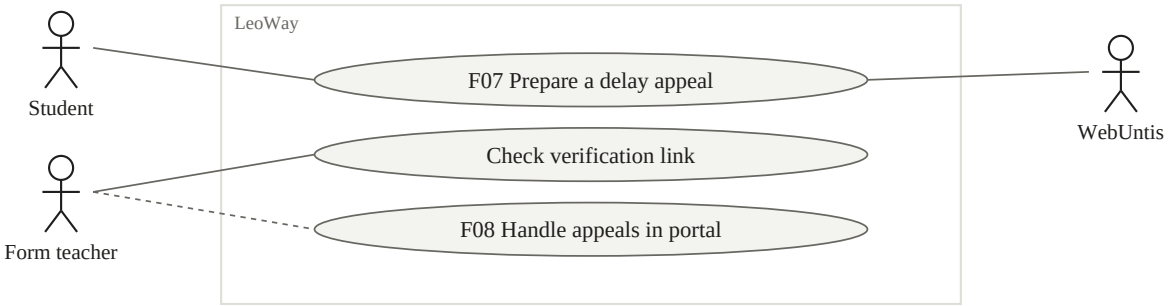
Standard scenario. The student writes *notify when delay > 5 min and first lesson is exam*. LeoWay checks the rule, reads exam days from WebUntis and alerts when both conditions are true.

Non-standard scenario. An invalid rule is rejected with the error position. If WebUntis has no exam data, exam days can be marked by hand.



2.3 Block C: Delay appeals

LeoWay matches lateness entries with delay records; the form teacher checks the appeal by link or in the optional portal.

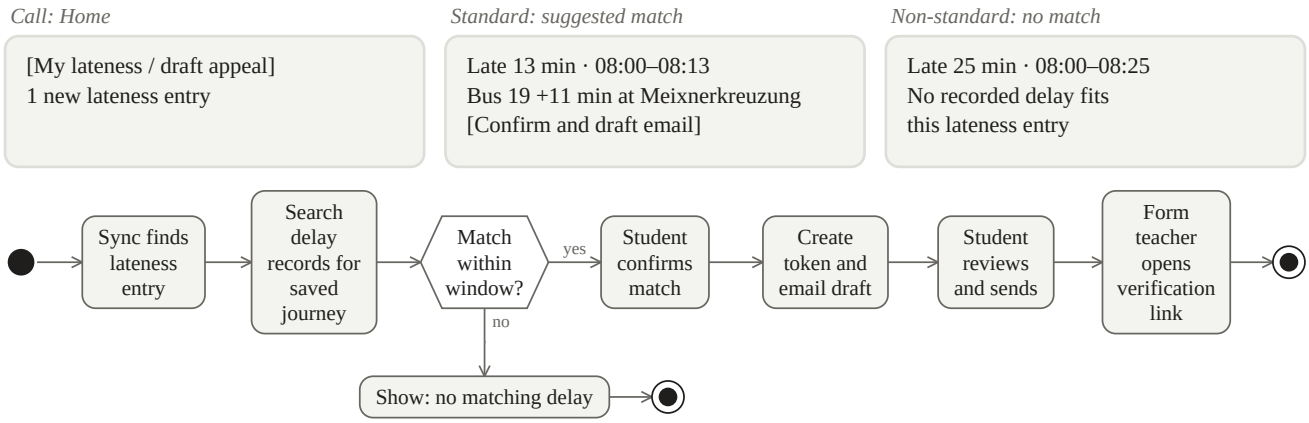


F07 · Prepare a delay appeal

Business process	BP2 Excusing lateness	**Goal**	The student sends a checkable appeal in under one minute; the form teacher can verify it.
Precondition	A lateness entry was synced from WebUntis; a delay record exists for the saved journey.	**Postcondition**	An appeal with a verification token exists; the email is opened in the student's mail app.
Involved roles	Student; form teacher; WebUntis	**Triggering event**	The noon or afternoon sync finds a new lateness entry.

Standard scenario. LeoWay shows "You were marked 13 min late; your bus 19 was 11 min late." The student confirms, checks the editable email (line, stop, times, minutes late, verification link) and sends it. The form teacher opens the link and sees only the delay record.

Non-standard scenario. No record fits the window (proposed: 15 minutes) or the absence is longer than the first lesson: no appeal is suggested. Changed, expired or revoked links show "Link not valid".

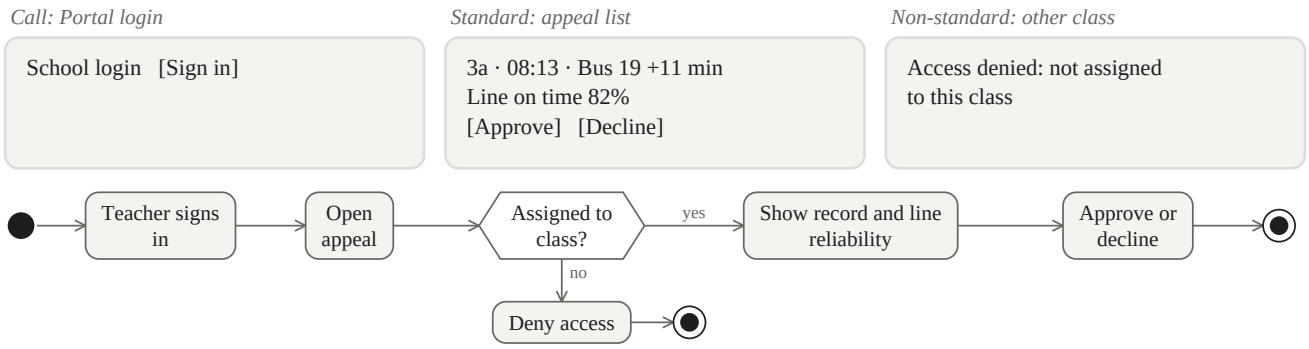


F08 · Handle appeals in the teacher portal (optional)

Business process	BP2 Excusing lateness	**Goal**	Enrolled form teachers handle appeals in one list instead of by email.
Precondition	The teacher is enrolled with a school-approved login and assigned to the class.	**Postcondition**	The appeal is marked approved or declined. The teacher still records the decision in WebUntis.
Involved roles	Form teacher	**Triggering event**	A student of the teacher's class sends an appeal.

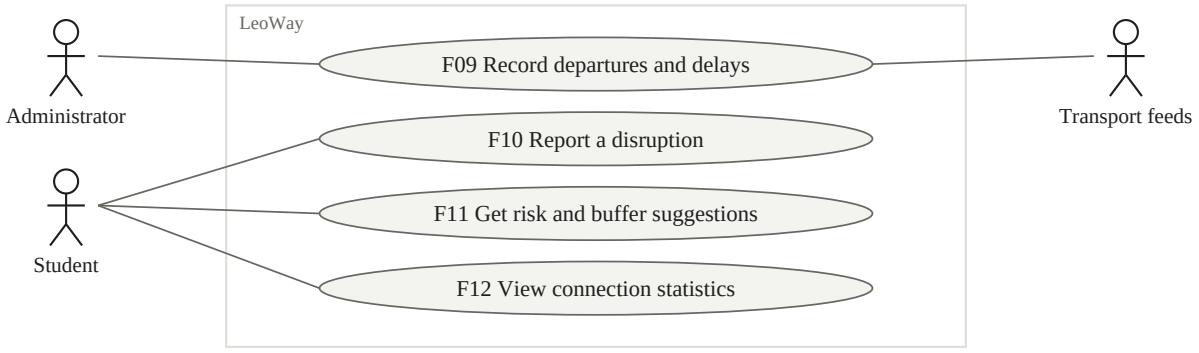
Standard scenario. The teacher opens the portal, sees the appeal with the line's reliability next to it and approves it.

Non-standard scenario. The teacher tries to open an appeal from another class. The portal denies access. Teachers who do not enrol keep receiving emails.



2.4 Block D: Delay data and community

Recorded delays feed warnings, appeals, prediction and statistics; students add crowd reports.

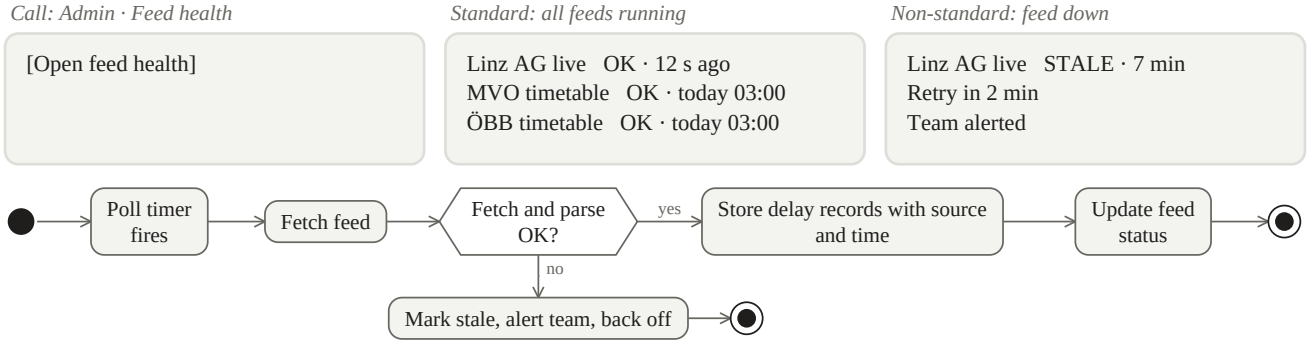


F09 · Record departures and delays

Business process	BP2 Excusing lateness, BP3 Reporting problem connections	**Goal**	A complete delay history with source and collection time.
Precondition	Feed adapters are configured.	**Postcondition**	New delay records are stored; the feed status shows the last update.
Involved roles	Administrator; transport feeds	**Triggering event**	The poll timer of a feed fires (also on holidays).

Standard scenario. The recorder fetches Linz AG departures, stores each new departure with line, trip, stop, bay, service date, planned time, delay, trip status, source and collection time, adds a new record when a departure's delay or trip status changes, and updates the feed status.

Non-standard scenario. A feed fails: other feeds keep running, the feed is marked stale after five minutes, the team is alerted and the recorder backs off.

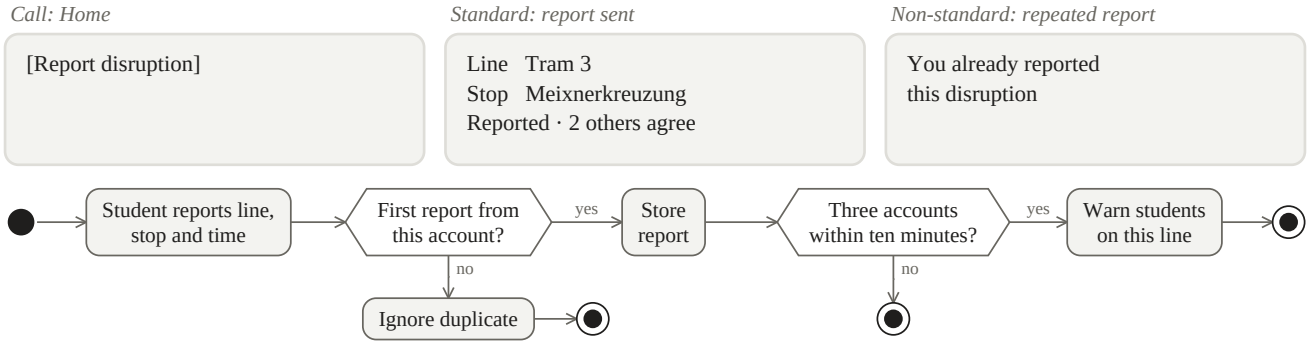


F10 · Report a disruption

Business process	BP1 Getting to school	**Goal**	Students warn each other before official feeds show a disruption.
Precondition	The student is logged in.	**Postcondition**	The report is stored; if the threshold is reached, a warning goes to students using that line.
Involved roles	Student	**Triggering event**	The student taps "Report disruption".

Standard scenario. Three accounts report tram 3 at Meixnerkreuzung within ten minutes (proposed). LeoWay warns students on that line, marked as a crowd report.

Non-standard scenario. One student reports the same disruption several times. The reports count once and no warning is sent.

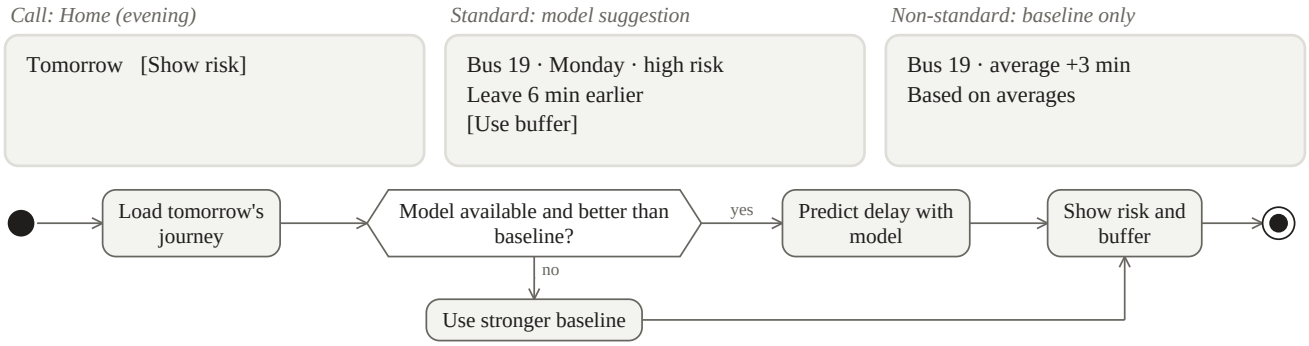


F11 · Get risk and buffer suggestions

Business process	BP1 Getting to school	**Goal**	The student sees how risky tomorrow's connection is and a personal buffer.
Precondition	Enough delay history exists for the connection (proposed: four weeks).	**Postcondition**	Home shows a risk level and a suggested buffer.
Involved roles	Student	**Triggering event**	The evening sync calculates the next day's journey.

Standard scenario. LeoWay shows "Bus 19 on Mondays: high risk, leave 6 minutes earlier." The student can accept the buffer.

Non-standard scenario. The model is unavailable or worse than the baseline on recent data. LeoWay uses the stronger baseline and labels the suggestion as based on averages.



F12 · View connection statistics

Business process	BP3 Reporting problem connections	**Goal**	The student council gets evidence about connections that arrive after lessons start.
Precondition	Delay records exist for the chosen period.	**Postcondition**	A report is shown and can be exported; it names no students.
Involved roles	Student (student council member)	**Triggering event**	The user opens "Statistics".

Standard scenario. The user picks the last school month. LeoWay lists connections ranked by how often they arrived after lessons started.

Non-standard scenario. A filter would leave fewer than five students behind a figure (proposed). LeoWay hides that figure so no student can be identified.

Call: Menu

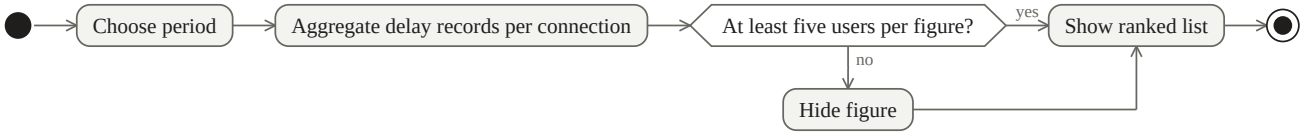
[Statistics]

Standard: late connections

September 2026
Bus 19 07:31 late 34%
Tram 3 07:25 late 12%

Non-standard: too few users

Bus 618 07:02
Hidden: fewer than 5 users



3. Non-functional requirements

Proposed targets on the agreed test server, checked in section 6.

3.1 Usability

Target group: students aged 14 to 19 checking their phone on the way to school, and form teachers who use email and WebUntis daily.

ID	Requirement
U01	The home screen works like the departure board at the stop: the first lesson, the leave-by time and the current delay of the student's own line are visible without scrolling on a 360 px wide phone screen.
U02	Preparing an appeal from a suggested match takes at most three taps until the email draft opens.
U03	Every time shown that comes from a prediction or a timetable (not live data) is labelled as such.
U04	The email draft reads like a normal message from the student; the teacher needs no LeoWay account to check the link.
U05	The website works with keyboard only and meets WCAG 2.1 AA contrast.
U06	Proposed support: current Chrome, Firefox and Safari, and Android 10 or later for the Capacitor app.

3.2 Efficiency

ID	Requirement
E01	Support 100 accounts and 20 active users at once. At least 95% of cached API requests finish within two seconds.
E02	Evaluate a received delay within 60 seconds. Send only one alert per channel for the same change.
E03	Live data older than five minutes is marked as stale.
E04	Find a leave-by time or alternative connection within one second for a saved route.

3.3 Maintainability

LeoWay runs over two school years and may later move to other schools or another team.

ID	Requirement
M01	Each feed sits behind one adapter interface. Adding a feed or changing one does not change the recorder or other adapters.
M02	Alert channels share one notification interface; a new chat platform is added as one new implementation.
M03	Adapter, timetable, matching and permission tests run in CI on every pull request. Code is reviewed before merge.
M04	The repository contains setup and deployment steps, the API description (OpenAPI) and this specification.

3.4 Security

ID	Requirement
S01	**Confidentiality:** HTTPS only. WebUntis keys are encrypted; the encryption key is kept outside the database and repository. Keys and private school data are never logged.
S02	**Confidentiality:** the API checks record ownership on every private request. The administrator role has health and feed endpoints only. Database support access uses a separate, restricted account and is logged.
S03	**Confidentiality:** verification links use random, unguessable tokens. Proposed: they expire after 30 days and can be revoked. They show only the delay record, source and collection time.
S04	**Data integrity:** delay records cannot be changed after collection. Students can only confirm a match, not edit times. Invalid links cannot read or change records.
S05	**Data integrity:** the WebUntis adapter calls only login/logout, timetable, timegrid, absence and exam endpoints. Messages, homework, grades and other data are blocked.
S06	**Availability:** a failed feed does not stop other feeds. Proposed: the recorder reaches 99% uptime on school days between 05:00 and 10:00. Encrypted daily backups are kept for seven days, with a restore test before release.

3.5 Legal constraints

ID	Requirement
L01	GDPR and the Austrian Data Protection Act apply. Use is voluntary; students aged 14 or older consent themselves. Privacy information is shown before connecting WebUntis.
L02	Data minimisation: no GPS, no home address and no WebUntis password. Absence reasons and notes are never stored. Deleted accounts stay deleted after a restore.
L03	Retention: routes and contacts until account deletion; timetable caches are replaced on sync; appeals and linked lateness entries until school-year end or account deletion.
L04	The school (school IT) agrees to the use of student access keys and the sync frequency before real student data is used.
L05	Data licences are respected: ÖBB timetables are CC BY 4.0 and credited in the app. VAO stays off until education access and limits are agreed; the stored Start contract allows only 100 queries per day per service.
L06	LeoWay provides evidence only. Excusing lateness stays the form teacher's decision under school rules.

4. Quantity structure

First estimates for 100 accounts (design limit 1,500), checked against measurements in the first sprints.

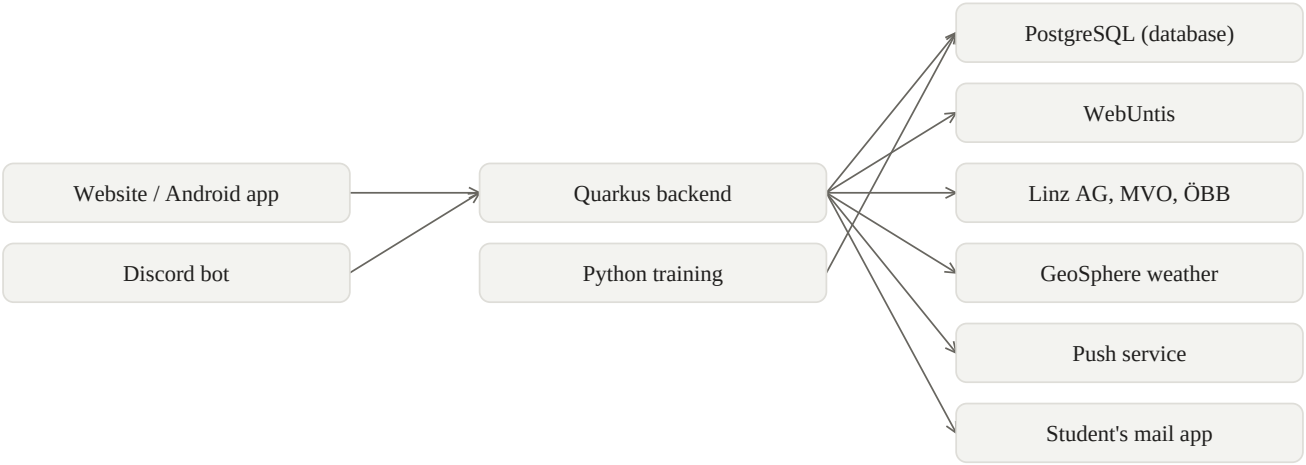
Quantity	Estimate	Basis
Student accounts	100 planned, 1,500 design limit	Efficiency target E01
Saved routes	1–2 per student, up to 3,000	Morning route and way home
Stops (master data)	about 10,000	Imported from MVO and ÖBB GTFS; replaced on each import
Watched stops for live data	about 50	Stops on saved routes plus the two school stops
Delay records per day	about 60,000	50 stops × about 300 departures per day × about 4 changes each. Measured in Sprint 1 (A09): about 2,700 records per school day at the two school stops
Delay record storage	about 0.23 KB each, about 5 GB per year	Including indexes (measured in A09); history is kept for prediction

Quantity	Estimate	Basis
WebUntis requests per day	about 3,000 at 100 accounts	About 30 sync calls per student per school day (section 5.2)
Lateness entries per day	up to 20 at 100 accounts	Assumption: a few students are late on a bad day
Appeals per day	up to 20	At most one per lateness entry
Crowd reports per day	up to 50	Mostly on disruption days
Alerts per day	up to 600	100 students × 2 alerts × up to 3 channels
Data per alert	under 2 KB	Short text message
Data per app start	under 50 KB	Home data without static files

5. System architecture and interfaces

5.1 Components

Angular provides the website and teacher portal; Capacitor wraps it as the Android app. Quarkus runs the backend with PostgreSQL. Python trains the prediction model on the team's hardware. Each source has its own adapter; all channels share one alert ID to avoid duplicates.



5.2 Interfaces

Partner	Interface
Website and Android app	REST/JSON over HTTPS to the Quarkus backend; push data messages to the Android app for alarm changes.
WebUntis	JSON-RPC with the student's access key (webuntis package), read only. Allowed calls: login/logout, timetable, timegrid, absences and exams (getExamsForRange).
Linz AG Linien	Live departures and delays (GTFS-RT, EFA departure endpoint), polled by the recorder.
MVO, ÖBB	GTFS timetables, imported nightly.
VAO	Off until education access and limits are agreed. Without live data, show "timetable only".
GeoSphere Austria	Weather data as input for prediction.
Discord, push, email	One notification interface; each channel is one implementation. Appeals open as a mailto draft in the student's own mail app.
Form teacher	Verification link in the browser; optional portal with school-approved login.

WebUntis sync (proposed, Europe/Vienna): lessons and exams at 20:00 for the next day, at 05:00 and every 15 minutes from 05:00 to 10:00; absences at 12:00 and 16:00. Students can refresh manually. Data is cached and requests back off after errors; limits are confirmed with school IT.

Alarm updates: after a lesson change, the backend pushes the new alarm time with a version to the phone. The app cancels the old alarm and sets the new one, and ignores older versions. It syncs again when opened. If updates cannot arrive, it shows the last sync time; alarm changes are not guaranteed while offline or force-stopped.

5.3 Domain model



5.4 Calculation and time rules

Choose a journey that arrives before the first lesson, allowing the school walk and buffer. Subtract the walk to the stop from the departure time. If none fits, show a warning.

Keep the trip's service date. Store timestamps in UTC and display Europe/Vienna time. Mark predicted times as predictions. Proposed: suggest a match if the saved journey's school arrival is within 15 minutes of the end of the lateness entry. The student confirms every match.

6. Acceptance criteria

Tests follow the standard and non-standard scenarios in section 2. The teacher and team agree which features are delivered; each passes on its own. Private records use test data.

Test	Pass condition
A01 · F01, S05	Standard: connect with a valid key; an exam is found without manual marking; the noon or afternoon sync finds a new lateness entry. Non-standard: a wrong key is rejected and nothing is stored; forbidden data calls are blocked.
A02 · F02	Standard: a leave-by time is shown for a saved route. Non-standard: cancel the first lesson overnight and in the morning; the leave-by time and phone alarm move. Ignore an older push; check offline recovery and a daylight-saving change.
A03 · F03	Standard: a delay that causes late arrival shows a warning and an alternative, plus the exam warning when needed. Non-standard: without live data, "timetable only" is shown.
A04 · F04, E02	Standard: one message per chosen channel within 60 seconds; the alarm rings on a real Android phone. Non-standard: with permission denied, a normal notification is offered.
A05 · F05	Standard: the way home follows the last lesson. Non-standard: end lessons early; the next connection after the new end time is shown.
A06 · F06	Standard: a delay/exam rule fires using a WebUntis exam. Non-standard: an invalid rule shows an error; manual marking works when exam data is missing.
A07 · F07, U02	Standard: three students each prepare the correct draft in under one minute; its link shows the chosen record. Non-standard: no match is suggested outside the window or for longer absences; changed, expired and revoked tokens are rejected.

Test	Pass condition
A08 · F08	If built: standard, a teacher handles an assigned appeal; non-standard, the teacher cannot open another class.
A09 · F09, E03, S06	Standard: records keep source and time. Non-standard: disconnect a feed; old data is marked stale and other feeds keep working.
A10 · F10	Standard: three accounts trigger a warning. Non-standard: repeated reports from one account count once.
A11 · F11	Test on later dates excluded from training. Compare mean delay error with both baselines. Non-standard: keep the stronger baseline if the model loses.
A12 · F12	Standard: a known late connection appears in the statistics. Non-standard: figures with fewer than five users are hidden.
A13 · S01–S04, L02	Student A cannot read B's records. An administrator cannot use private endpoints. Deletion removes private data and survives a backup restore.
A14 · E01, E04, U01, U05, U06, M03	Meet the load targets, phone-width and keyboard checks in the supported browsers, a test of the Android app on a real phone, and CI checks; record the results.

6.1 Delivery and open decisions

Deliver code, database changes, tests, deployment steps, a short user guide and test results. Work in two-week sprints. Update the backlog and this document when requirements change.

Confirm the final name, product owner, hosting, WebUntis permission and limits, VAO access, the teacher email source and the optional teacher login. Until email lookup is agreed, the student enters and checks the address. Review proposed settings with the teacher before release.